Project: Worked examples
Title: Thin cylindrical shell under internal pressure
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Minimum shell thickness of a thin cylindrical pressure vessel under internal pressure for circumferential and longitudinal stress, the governing required thickness with corrosion allowance and design factor, the hoop stress at the corroded thickness provided, and the provided thickness against the requirement.

WHAT IT CALCULATES	RELATION
Thickness needed for the circumferential stress	$t_c = P \cdot R/(S \cdot E - 0.6 \cdot P)$
Thickness needed for the longitudinal stress	$t_l = P \cdot R/(2 \cdot S \cdot E + 0.4 \cdot P)$
Required thickness, with the design factor and corrosion allowance	$t_{req} = n \cdot \max(t_c, t_l) + c$
Hoop stress in the plate provided, after corrosion	$\sigma_\theta = P \cdot (R + 0.6 \cdot t)/(E \cdot t)$, at $t = t_p - c$
Range of validity of the thin-wall formulae	$t_c \le R/2$ and $P \le 0.385 \cdot S \cdot E$

Basis: Thickness of shells under internal pressure, circumferential stress[1](UG-27(c)(1)); Thickness of shells under internal pressure, longitudinal stress[1](UG-27(c)(2)).

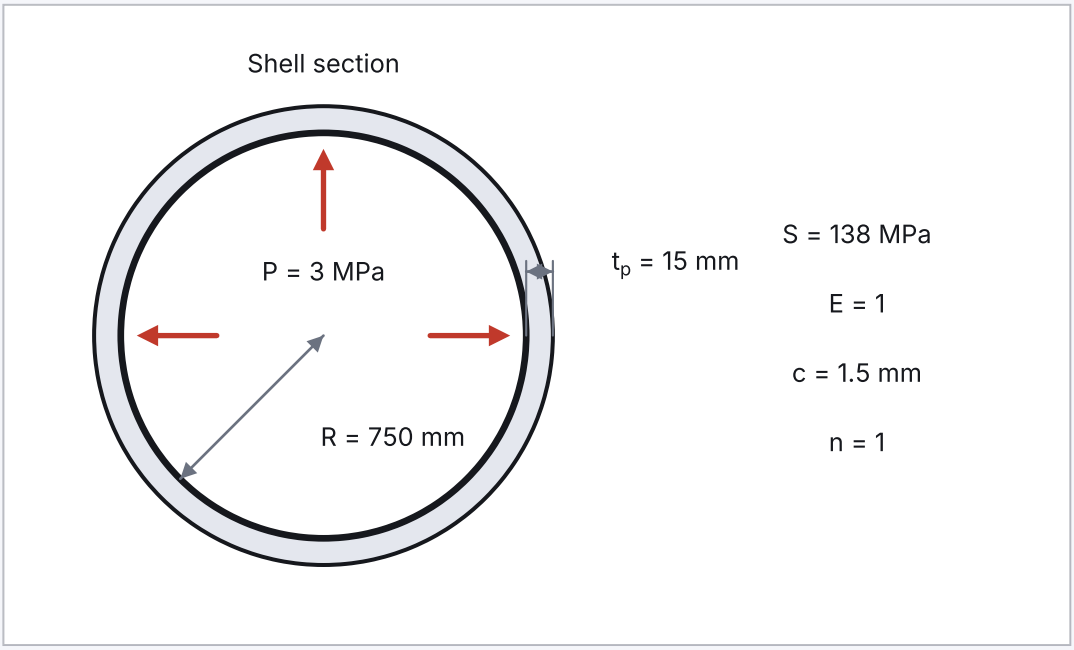


Figure 1 — A thin-walled cylinder in section under internal pressure: inside radius R, shell thickness t_p including the corrosion allowance c. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
P	Internal design pressure	3	MPa	spec: LOAD-SCHEDULE-01	0.01 … 100 MPa

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
R	Inside radius (corroded)	750	mm	drawing: DRG-PV-602 rev A	5 … 10000 mm
S	Allowable stress at design temperature	138	MPa	spec: SPEC-MECH-03 design criteria	10 … 1000 MPa
E_j	Joint efficiency	1		spec: SPEC-MECH-03 design criteria	0.45 … 1
n	Design factor on the computed thickness (legacy; 1 for none)	1		spec: SPEC-MECH-03 design criteria	1 … 5
c	Corrosion allowance	1.5	mm	spec: SPEC-MECH-03 design criteria	0 … 25 mm
t_p	Shell thickness provided (nominal, before corrosion)	15	mm	drawing: DRG-PV-602 rev A	0.5 … 500 mm

3. Assumptions

A1. Cylindrical shell away from heads, nozzles and supports; membrane stresses only. (UG-27 scope.)
A2. The joint efficiency reduces the allowable stress uniformly. (Code-form simplification carried from the legacy module.)

4. Working

[8] $t_c = P * R / (S * E_j - 0.6 * P)$ [1](VIII-1 UG-27(c)(1))
$= 3\ \text{MPa} \times 750\ \text{mm} / (138\ \text{MPa} \times 1 - 0.6 \times 3\ \text{MPa})$
$= 16.52\ \text{mm}$
Thickness for circumferential stress

[9] $t_l = P * R / (2 * S * E_j + 0.4 * P)$ [1](VIII-1 UG-27(c)(2))
$= 3\ \text{MPa} \times 750\ \text{mm} / (2 \times 138\ \text{MPa} \times 1 + 0.4 \times 3\ \text{MPa})$
$= 8.1169\ \text{mm}$
Thickness for longitudinal stress

[10] $t_{req} = n * \max(t_c, t_l) + c$
$= 1 \times \max(16.52\ \text{mm}, 8.1169\ \text{mm}) + 1.5\ \text{mm}$
$= 18.02\ \text{mm}$
Required nominal thickness (design factor and corrosion allowance included)

[11] $t_{corr} = t_p - c$
$= 15\ \text{mm} - 1.5\ \text{mm}$
$= 13.5\ \text{mm}$
Corroded thickness provided

[12] $\sigma_h = P * (R + 0.6 * t_{corr}) / (E_j * t_{corr})$ [1](VIII-1 UG-27(c)(1))
$= 3\ \text{MPa} \times (750\ \text{mm} + 0.6 \times 13.5\ \text{mm}) / (1 \times 13.5\ \text{mm})$
$= 168.47\ \text{MPa}$
Hoop stress at the corroded thickness provided

[13] $u = t_{req} / t_p$
$= 18.02\ \text{mm} / 15\ \text{mm}$
$= 1.2013$
Required over provided thickness

5. Results

SYMBOL	VALUE	DESCRIPTION
t_c	16.52 mm	Thickness, circumferential stress
t_l	8.117 mm	Thickness, longitudinal stress
t_{req}	18.02 mm	Required nominal thickness
σ_h	168.5 MPa	Hoop stress at the corroded thickness
u	1.21	Thickness utilisation (governing)

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
FAIL	Provided thickness against required	u = 1.21	u ≤ 1	1.21	[1](VIII-1 UG-27)

7. Applicability

None.

8. Verification

Template mech.thin-vessel@1.0.1 Content hash 89ebf5d7f4e4e4fb Engine calc-engine/1.0.0
Known-answer cases passed 2/2 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. American Society of Mechanical Engineers. Boiler and Pressure Vessel Code, Section VIII, Division 1: Rules for Construction of Pressure Vessels. ASME.